

Dynamics of butyl- and hexyl-methylimidazolium bromide ionic liquids

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We have carried out quasi-elastic neutron scattering measurements, supported by classical molecular dynamics simulations, to determine the liquid state dynamics of 1-butyl-3-methylimidazolium bromide and 1-hexyl-3-methylimidazolium bromide. The spectra are fitted with two- and three-Lorentzian models and indicate the existence of a localized dynamics of the alkyl chains in the sub-picosecond range. The spatial dependence of the dynamics is consistent with a confined translational diffusion model with parameters consistent with the molecular dynamics simulations.

1. Introduction

Room-temperature ionic liquids (RTILs) are receiving increased attention due to their unique properties, including non-volatility and solvating capability, leading to a wide range of potential uses in catalysis, separation technology, photovoltaics and fuel cells [1-12]. High freezing temperatures are generally achieved by combining large, asymmetrically substituted organic cations with either small or large anions.

The length of the nonpolar side chains in the cations provides a sensitive handle for tuning the structure and properties of an ionic liquid. Many studies in the past decade have focused on the effects of changing the lengths of the alkyl chains in imidazolium-based liquids [13]. Regarding physical properties, their general trend is for a decreasing mass density, molar atomic number density, self-diffusion coefficient and ionic conductivity, and increasing viscosity with increasing chain length [14].

We have carried out a systematic investigation of the ionic liquid series 1-alkyl-3-methylimidazolium bromide ionic liquids, $C_n\text{mimBr}$, with $n = 2, 4$ and 6 . We have previously reported high-energy x-ray diffraction (HEXRD) measurements and atomistic molecular dynamics (AMD) numerical simulations on all three liquids. For $C_2\text{mimBr}$, the liquid structure factor $S(Q)$ is dominated by an intense peak at a scattering vector $Q = 1.7 \text{ \AA}^{-1}$, associated mainly with the packing of the anions around the large cations [15], whereas in $C_4\text{mimBr}$ and more strikingly in $C_6\text{mimBr}$ a low- Q peak rises and moves to lower Q with increasing chain length, reaching $0.33 \text{ \AA}^{-1}$ in $C_6\text{mimBr}$ [16]. This feature is consistent with the concept of nanoscale heterogeneity with small, crystal-like moieties [17]. For all three liquids, excellent agreement between the HEXRD experiments and AMD simulations is obtained, including the region of the low- Q peak that has proved problematic in previous work in the

literature. The ring-ring, anion-anion and ring-anion partial structure factors are dominated by strong charge-ordering peaks around $1.1 \text{ \AA}^{-1}$, while the tail-tail $S(Q)$ is dominated by the low-Q peak [16].

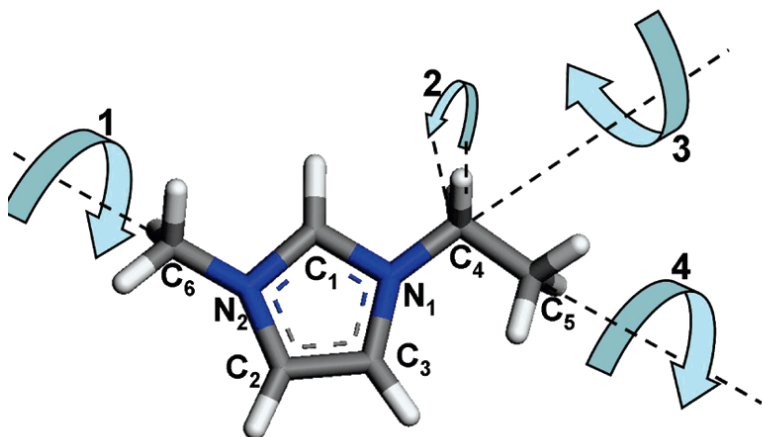


Fig. 1. Possible reorientational motions of the C_2mim^+ cation.

We have also studied the dynamics of the liquids with quasielastic neutron scattering (QENS) experiments performed in parallel with the AMD simulations. Results in this domain can be connected with important macroscopic parameters such as diffusion constants and viscosities and furthermore lend themselves to parallel numerical simulations with AMD, which turns out to be essential for interpreting experimental results on these complex systems. However, the number of QENS studies is still relatively small [18-22]. For C_2mimBr , we have studied the dynamics of both crystal and liquid phases [23]. Figure 1 shows the structure of the C_2mim^+ cation with the possible reorientation motions, denoted by labels 1-4. According to the MD simulations, the global reorientations (1) are too slow to give rise to perceptible broadening with the resolution of the QENS measurements; this is confirmed by recent NMR measurements [24] that showed that the relaxation time for the rotation of the full C_2mimBr molecule is longer than 1 ns. On the other hand, the CH_2 group dynamics (2) and methyl group motions (4) are too fast, so that the broadened quasielastic line would look like an incoherent background. Thus, the ethyl group reorientations (3) represent the only possible contribution to the observed quasielastic broadening. Both experiment and simulation demonstrated that, in the length and time scales being probed (fractions of a nm and a few ps), the dynamics are dominated by activated translational diffusion in the liquid phase and reorientations of the ethyl groups in both solid and liquid.

2. Results and Discussion

In this report we extend our dynamical studies to C_4mimBr and C_6mimBr , which for convenience we refer to as $BmimBr$ and $HmimBr$, respectively. QENS measurements were performed on two neutron time-of-flight spectrometers: NEAT at the Berlin Neutron Scattering Center and IN5 at the ILL. The incident neutron wavelength in both instruments was $5.0 \text{ \AA}$, giving an energy resolution of $^{0.1} \text{ meV}$ (FWHM). A representative set of data at 392 K is shown in Fig 2. In both liquids a broadened quasi-elastic line is clearly seen, with a width that, as expected, increases with Q and with temperature (not shown here). There are also slow dynamics with a time scale of over $\sim 10 \text{ ps}$ that are not resolved and contribute to a strong central elastic peak.

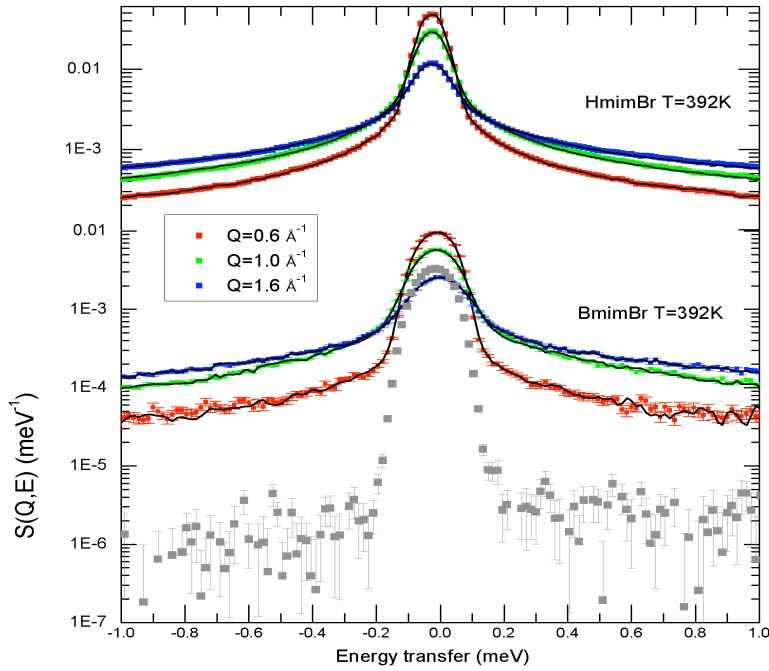


Fig. 2. Measured QENS scattering functions $S(Q,E)$ (E = energy transfer) at three Q values for BmimBr and HmimBr at 392 K, plotted with a logarithmic scale on the ordinate axis. The grey points refer to the measured resolution function. The solid curves represent the fits described in the text.

As in the case of C_2 mimBr, the BmimBr scattering function could be satisfactorily fitted by a sum of two Lorentzians that can be associated with translational and reorientation relaxation:

$$S(Q,E) = \exp(-Q^2\langle u^2 \rangle) \{A_0(Q)L(W_t, E) + [1 - A_0(Q)]L(W_{t+r}, E)\} \quad (1)$$

where $\langle u^2 \rangle$ is the mean-square amplitude of the vibrations (with energies assumed to be outside the QENS window), $A_0(Q)$ is the form factor, $L(W,E)$ is the Lorentzian function of unit area with W its full-width at half-maximum (FWHM), W_t is the FWHM of the narrow Lorentzian translational component and W_r is the FWHM of the broader Lorentzian of the reorientation component.

For HmimBr an additional localized relaxation mechanism evidently comes into play due to the longer floppy alkyl chain and additional Lorentzians are needed to describe the scattering function:

$$S(Q,E) = \exp(-Q^2\langle u^2 \rangle) \{A_{0,2}(Q)A_{0,3}(Q)L(W_t, E) + A_{0,3}(Q)[1 - A_{0,2}(Q)]L(W_{t+r_2}, E) + A_{0,2}(Q)[1 - A_{0,3}(Q)]L(W_{t+r_3}, E) + [1 - A_{0,2}(Q)][1 - A_{0,3}(Q)]L(W_{t+r_2+r_3}, E)\} \quad (2)$$

where W_{r_2} , $A_{0,2}$ and W_{r_3} , $A_{0,3}$ are respectively the FWHM and form factors corresponding to the two localized relaxation mechanisms.

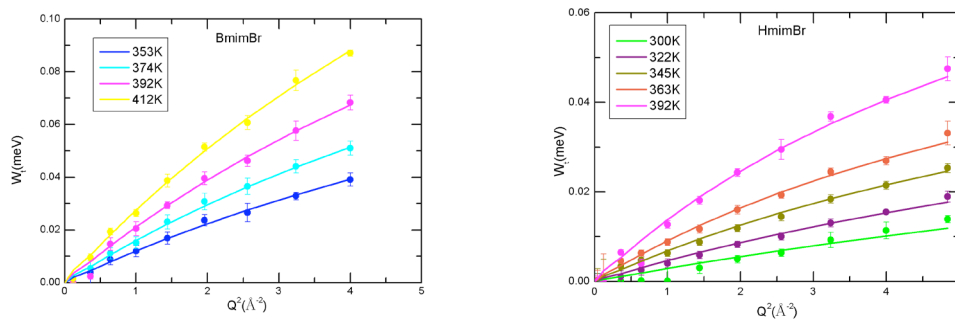


Fig. 3. Fitted values of the FWHM of the translational component for BmimBr and HmimBr at several temperatures.

The fits according to Eqs. (1) and (2), convolved with the measured resolution function, are shown by the solid curves in Fig. 2. The FWHM of the translational can be described by the rapid jump diffusion model [25]:

$$W_t = \frac{2\hbar D Q^2}{1 + D Q^2 \tau_0} \quad (3)$$

where D is the diffusion constant and τ_0 is a characteristic residence time between jumps. Figure. 3 shows the values of W_t for BmimBr and HmimBr liquids obtained from the fits of Eqs. (1) and (2). Even though these are smaller than the instrumental resolution we consider them reliable due to the good statistics for the narrow component. The fits of Eq. (3) at the whole set of measured temperatures are also shown in Fig. 3, and the fitted values of D and τ_0 are given in Table I. Comparing the three liquids $C_2\text{mimBr}$ [23], BmimBr and HmimBr, the diffusion constant decreases and residence time increases with increasing alkyl chain length.

Table I: Values of the translational diffusion constant D and residence time τ_0 of BmimBr and HmimBr

BmimBr			HmimBr		
T (K)	D (10^{-10} m ² /s)	τ_0 (ps)	T (K)	D (10^{-10} m ² /s)	τ_0 (ps)
353	1.98	4.17	300	0.46	10.91
373	2.66	3.40	322	0.76	9.90
392	3.51	2.64	345	1.15	8.75
412	4.58	2.01	363	1.56	7.88
			392	2.38	5.72

The broader Lorentzians represents the faster and the localized dynamics. Figure 4 shows the obtained values for the W_r for BmimBr and HmimBr liquids; a gradual increase with Q is observed, due to the unavoidable coupling of translational and localized modes in a liquid. Extrapolation to $Q = 0$ gives values of the localized relaxation times. For BmimBr the single rotational relaxation time is $\tau_R \approx 1.5$ ps and for HmimBr the two relaxation times are $\tau_{R2} \approx 2.2$ ps and $\tau_{R3} \approx 8.8$ ps.

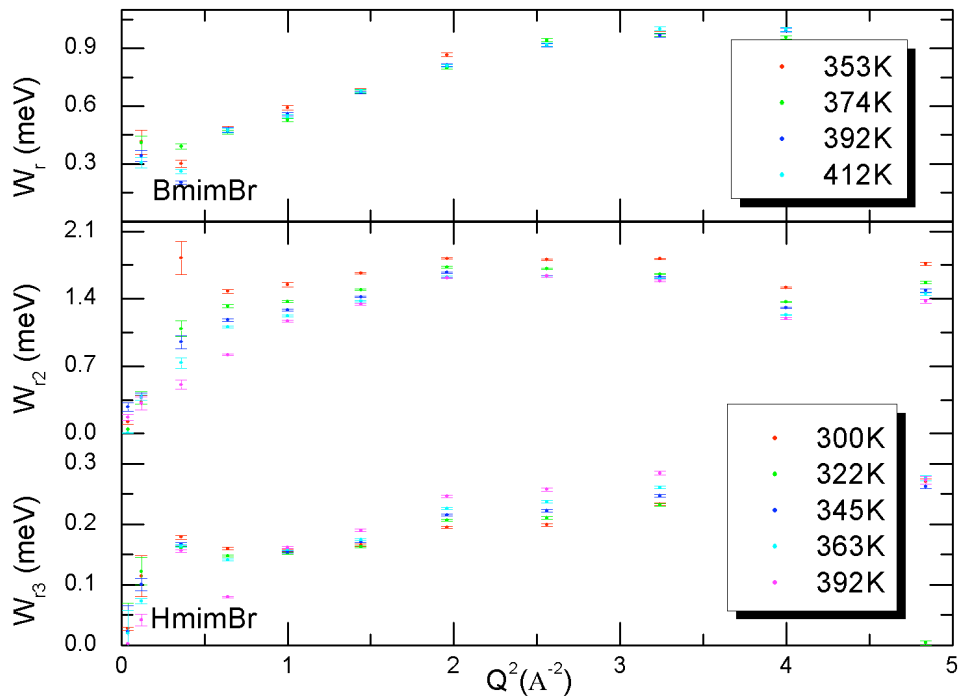


Fig. 4. Fitted values of the FWHMs of the localized components for BmimBr and HmimBr at several temperatures.

The fitted values of $A_0(Q)$ for BmimBr are shown in Fig. 5 and those of $A_{0,2}(Q)$ and $A_{0,3}(Q)$ for HmimBr in Fig. 6. All show a monotonous decreasing behavior with Q and attenuate more rapidly at higher temperatures due to the more extended motions.

In the MD studies of both liquid and solid C_2mimBr , it was found that the dynamics consist essentially of jumps between two pairs of inequivalent sites in the solid and jumps between four equivalent sites in the liquid, and contour plots of the average spatial distribution of C5, the end carbon atom of the ethyl group, relative to the N1 nitrogen provided the basis for a model for fitting the QENS data. With the more complex dynamics in BmimBr and HmimBr due to the longer alkyl chains, we were not able make such identification and had recourse to a phenomenological model. After trying several different models the best results were obtained with the confined translational diffusion model of Dianoux and Volino [26]:

$$A_0(Q) = c + (1 - c) \left(\frac{3j_1(Qa)}{Qa} \right)^2 \quad (4)$$

where j_1 is the first-order spherical Bessel function, the parameter c describes the immobile fraction of hydrogen atoms in the fast relaxation processes and the parameter a defines the confinement sphere radius of the participating hydrogen atoms. This two-parameter form gives reasonable fits to all three form factors, as shown in Figs. 5 and 6. The values of the c and a parameters are listed in Table II.

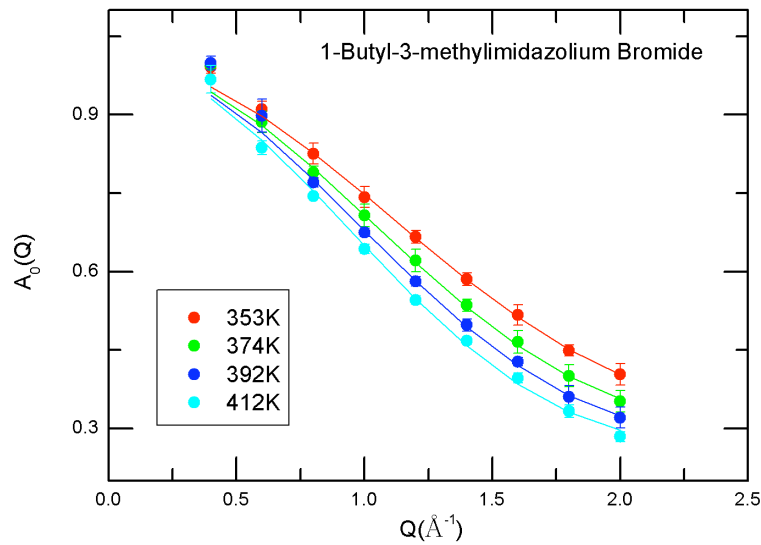


Fig. 5. Fitted values of the form factor for BmimBr at the temperatures measured.

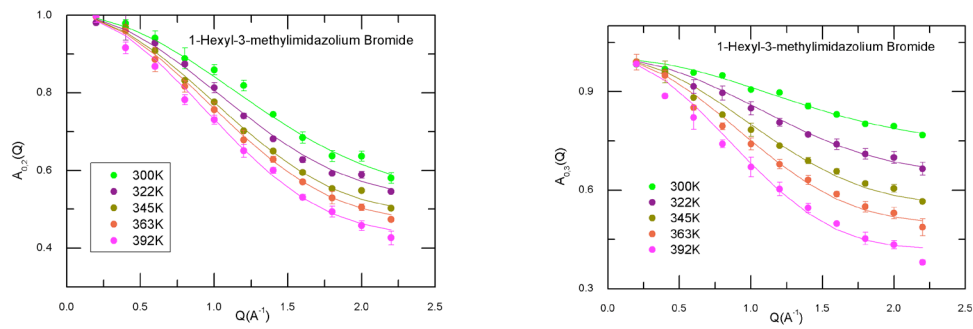


Fig. 6. Fitted values of the two form factors for HmimBr at the temperatures measured.

For BmimBr, the total number of hydrogen atoms is 15 where only the 7 ones of the last three carbon atoms of the butyl group are expected to be mobile. Therefore the ratio of mobile hydrogen atoms $7/15 \approx 0.46$. This is higher than the fitted values unless we only count the hydrogen atoms of the last 2 carbons ($5/15 \approx 0.33$). For HmimBr, the total number of hydrogen atoms is 19. The hexyl chain is two carbon atoms longer than the butyl one, and the hexyl chain is more flexible and even its very first atoms tend to bend. Two situations can be imagined, with either the whole chain ($13/19 \approx 0.68$) or only the last part of the chain (the last 5 carbon atoms) contributing to the relaxation process ($11/19 \approx 0.57$). Both situations are in reasonable agreement with the fitted values of the c parameter of the two form factors.

Table II: Values of the confined translational diffusion model fitted to the form factors for BmimBr and HmimBr.

Parameters					
BmimBr $A_0(Q)$	T (K)	c	Error(c)	a (Å)	Error(a) (Å)
	353	0.32	0.004	1.50	0.009
	373	0.30	0.003	1.61	0.001

	392	0.28	0.003	1.68	0.007
	412	0.27	0.004	1.76	0.007
HmimBr $A_{0,2}(Q)$	T (K)	c_2	Error (c_2)	a_2 (Å)	Error (a_2) (Å)
	300	0.55	0.003	1.45	0.010
	322	0.54	0.002	1.65	0.006
	345	0.50	0.001	1.71	0.09
	363	0.49	0.001	1.72	0.10
	392	0.44	0.002	1.76	0.007
HmimBr $A_{0,3}(Q)$	T (K)	c_3	Error (c_3)	a_3 (Å)	Error (a_3) (Å)
	300	0.75	0.002	1.44	0.01
	322	0.68	0.001	1.71	0.007
	345	0.61	0.001	1.95	0.006
	363	0.54	0.001	2.00	0.006
	392	0.45	0.002	2.05	0.007

According to the MD simulations, the global reorientation of the cations is too slow to give rise to perceptible broadening in the QENS measurements. Figure. 7 shows the possible positions of a hydrogen atom of the hexyl chain's methyl group in the Hmim⁺ cation. The edge of the density map (blue dots) is behind the ring. That means that the hexyl chain can fold significantly and wrap around the ring, which is not the case for the butyl chain of the Bmim⁺ cation. The same representation can be performed for the other hydrogen atoms of the alkyl chain. Hence one can imagine spherical shells, each a few Å thick and embedded in each other. The hydrogen atoms are moving in a confined space, more or less spherically shaped, as described by the confined translational diffusion model.

3. Conclusion

In conclusion, the dynamics of liquid BmimBr and HmimBr exhibit translational motions and one and two localized processes. The spatial dependence of the dynamics is consistent with a confined translational diffusion model with parameters consistent with the molecular dynamics simulations.

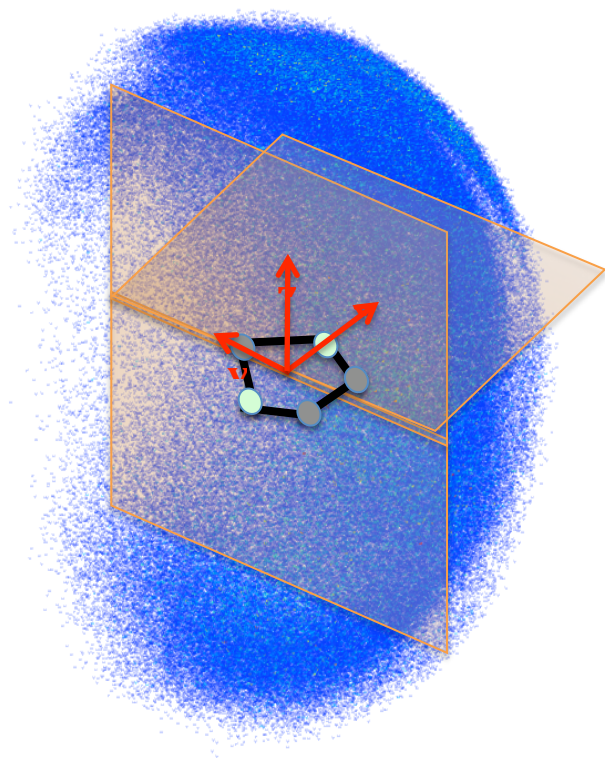


Fig 7. Contour plots of the average spatial distribution of the hydrogen atoms of the hexyl chain's methyl group in the Hmim⁺ cation.

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References

- ¹ M. Armand, *Solid State Ionics* **69**, 309 (1994)
- ² M. L. Dietz, *Sep. Sci. Technol.* **41**, 2047 (2006)
- ³ X. Han and D. W. Armstrong, *Acc. Chem. Res.* **40**, 1079 (2007)
- ⁴ P. Wasserscheid and T. Welton, Eds., *Ionic Liquids in Synthesis* (Wiley-VCH: New York, 2003)
- ⁵ C. A. Angell, N. Byrne and J.-P. Belieres, *Acc. Chem. Res.* **40**, 1228 (2007)
- ⁶ M. Armand, F. Endres, D. R. MacFarlane, H. Ohno and B. Scrosati, *Nat. Mater.* **8**, 621 (2009)
- ⁷ P. Simon and Y. Gogotsi, *Nat. Mater.* **7**, 845 (2008)
- ⁸ T. Welton, *Chem. Rev.* **99**, 2071 (1999)
- ⁹ F. Van Rantwijk and R. A. Sheldon, *Chem. Rev.* **107**, 2757 (2007)
- ¹⁰ M. Smiglak, A. Metlen and R. D. Rogers, *Acc. Chem. Res.* **40**, 1182 (2007)

- ¹¹ M. P. Stracke, G. Ebeling, , R. Cataluña and J. Dupont, *Energy Fuels* **21**, 1695 (2007)
- ¹² A. Doroodian, J. E. Dengler, A. Genest, N. Rösch and B. Rieger, *Angew. Chem. Int. Ed.* **49**, 1871 (2010)
- ¹³ J. de Andrade, E. S. Boes and H. Stassen, *J. Phys. Chem. B* **113**, 7541 (2009)
- ¹⁴ A. A. Fannin, D. A. Floreani, L. A. King *et al.*, *J. Phys. Chem.* **88**, 2614 (1984)
- ¹⁵ B. Aoun, A. Goldbach, S. Kohara, J.-F. Wax, M. A. González, and M.-L. Saboungi, *J. Phys. Chem. B* **114**, 12623 (2010).
- ¹⁶ B. Aoun, A. Goldbach, M. A. González, S. Kohara, D. L. Price and M.-L. Saboungi, *J. Chem. Phys.* **134**, 104509 (2011).
- ¹⁷ H. V. R. Annapureddy, H. K. Kashyap, P. M. De Biase, and C. J. Margulis, *J. Phys. Chem. B* **114**, 16838 (2010).
- ¹⁸ A. Triolo, O. Russina, V. Arrighi, F. Juranyi, S. Janssen and C. M. Gordon, *J. Chem. Phys.* **119**, 8549 (2003)
- ¹⁹ A. Triolo, O. Russina, C. Hardacre, M. Nieuwenhuyzen, M. A. González and H. Grimm, *J. Phys. Chem. B* **109**, 22061 (2005)
- ²⁰ Y. Inamura, O. Yamamuro, S. Hayashi and H. Hamaguchi, *Physica B* **385-386**, 732 (2006)
- ²¹ E. Mamontov, H. M. Luo and S. Dai, *J. Phys. Chem. B* **113**, 159 (2009)
- ²² O. Russina, M. Beiner, C. Pappas, M. Russina, V. Arrighi, T. Unruh, C. L. Mullan, C. Hardacre and A. Triolo, *J. Phys. Chem. B* **113**, 8469 (2009)
- ²³ B. Aoun M. A. González, J. Olivier, M. Russina, Z. Izaola, D. L. Price and M.-L. Saboungi, *J. Phys. Chem. Lett.* **1**, 2503 (2010).
- ²⁴ M Imanari, K. Uchida, K. Miyano, H. Seki and K. Nishikawa, *Phys. Chem. Chem. Phys.* **12**, 2959 (2010).
- ²⁵ K. Singwi and A. Sjolander, *Phys. Rev.* **119**, 863 (1960)
- ²⁶ A. J. Dianoux and F. Volino, *Mol. Phys.* **34**, 1263 (1977); F. Volino and A. J. Dianoux, *Mol. Phys.* **36**, 389 (1978); **41**, 271 (1980).